Pareto Distribution

1 Definition

A continuous random variable X is said to follow a Pareto Distribution if its CDF is given by,

$$F_X(x) = \begin{cases} 0 & x < x_0, \quad x_0 > 0 \\ 1 - \left(\frac{x_0}{x}\right)^\theta & x > x_0, \quad \theta > 0 \end{cases}$$

PDF:

$$f_X(x) = \begin{cases} \frac{\theta x_0^\theta}{x^{\theta+1}} & \text{if } x \geqslant x_0 \\ 0 & \text{otherwise} \end{cases}$$

2 Moments

Existence of moments:

$$E|X|^n = \int_{-\infty}^{\infty} |x|^n f_X(x)dx = \int_{x_0}^{\infty} x^n \theta \frac{x_0^\theta}{x^{\theta+1}} dx = \theta x_0^\theta \int_{x_0}^{\infty} \frac{1}{x^{\theta-n+1}} dx,$$

which converges iff $\theta - n + 1 > 1 \Rightarrow$ iff $r < \theta$. Hence, $\mu'_r = E(X^r)$ exist iff $r < \theta$.
For $r < \theta$,

$$\begin{aligned} \mu'_r &= \int_{x_0}^{\infty} x^r \frac{\theta x_0^\theta}{x^\theta + 1} dx \\ &= \theta x_0^\theta \lim_{a\to\infty} \int_{x_0}^{a} \frac{dx}{x^{\theta-r+1}} \\ &= \theta x_0^\theta \lim_{a\to\infty} \left[\frac{x^{r-\theta}}{r-\theta} \Big|_{x_0}^{a} \right] \\ &= \theta x_0^\theta \lim_{a\to\infty} \left[\frac{a^{r-\theta} - x_0^{r-\theta}}{r-\theta} \right] \\ &= \theta x_0^\theta \lim_{a\to\infty} \left[\frac{x_0^{r-\theta} - a^{r-\theta}}{\theta - r} \right] \\ &= \theta x_0^\theta \frac{x_0^{r-\theta}}{\theta - r} \quad \text{as } \lim_{a\to\infty} a^{r-\theta} = 0 \\ &= \frac{\theta}{\theta - r} x_0^r \end{aligned}$$

$E(X) = \frac{\theta}{\theta - 1} x_0 \quad$ if $\theta > 1$.

$V(X) = \mu'_2 - {\mu'_1}^2 = \frac{\theta}{\theta - 2} x_0^2 - \left(\frac{\theta}{\theta - 1}\right)^2 x_0^2 = \theta x_0^2 \left(\frac{1}{\theta - 2} - \frac{\theta}{(\theta - 1)^2}\right) = \frac{\theta x_0^2}{(\theta - 2)(\theta - 1)^2}.$

$$\text{HM} = \frac{1}{H} = E\left(\frac{1}{X}\right) = \mu'_{-1} = \frac{\theta}{\theta+1}x_0^{-1}$$

$$\Rightarrow H = \left(1+\frac{1}{\theta}\right)x_0$$

GM:

$$\log G = E\left[\log_e X\right] = \int_{x_0}^{\infty} \ln x \frac{\theta x_0^{\theta}}{x^{\theta+1}} dx$$

As,

$$\int_{x_0}^{\infty} \frac{\theta}{x_0}\left(\frac{x_0}{x}\right)^{\theta+1} dx = 1$$

Differentiating w.r.t θ, $\int_{x_0}^{\theta}\left\{\frac{1}{x_0}\left(\frac{x_0}{x}\right)^{\theta+1} dx + \frac{\theta}{x_0}\left(\frac{x_0}{x}\right)^{\theta+1}\log\left(\frac{x_0}{x}\right)\right\} dx = 0$

$$\Rightarrow \frac{1}{\theta}\int_{x_0}^{\infty} f_X(x)dx + E\left[\log\left(\frac{x_0}{X}\right)\right] = 0$$

$$\Rightarrow \frac{1}{\theta} + E\left[\log\left(\frac{x_0}{X}\right)\right] = 0$$

$$\Rightarrow \frac{1}{\theta} + E\left[\log x_0 - \log X\right] = 0$$

$$\Rightarrow E[\log X] = \log x_0 + \frac{1}{\theta}$$

$\Rightarrow \log G = \log\left(x_0 e^{1/\theta}\right) \Rightarrow G = x_0 e^{1/\theta}$.

Verify that, AM>GM>HM
Now,

$$1+\frac{1}{\theta}+\left(\frac{1}{\theta}\right)^2+\left(\frac{1}{\theta}\right)^3+\cdots > 1+\frac{1}{\theta}+\frac{\left(\frac{1}{\theta}\right)^2}{2!}+\frac{\left(\frac{1}{6}\right)^3}{3!}+\cdots > 1+\frac{1}{\theta}$$

$$\Rightarrow \left(1-\frac{1}{\theta}\right)^{-1} > e^{1/\theta} > 1+\frac{1}{\theta}$$

$$\Rightarrow \frac{1}{1-\frac{1}{\theta}} > e^{1/\theta} > 1+\frac{1}{\theta}$$

$$\Rightarrow \frac{\theta}{\theta-1}x_0 > x_0 e^{1/\theta} > x_0\left(1+\frac{1}{\theta}\right)$$

$$\therefore \text{AM} > \text{GM} > \text{HM}$$

3 Median

$$\frac{1}{2} = F_X(me)$$

$$\Rightarrow \frac{1}{2} = 1-\left(\frac{x_0}{me}\right)^{\theta} \Rightarrow m_e = x_0 2^{1/\theta}$$

Remark

(1)Clearly, x_0 is the mode and the distribution is extremely positively skewed.

(2) $F_X(x) = 1-\left(\frac{x_0}{x}\right)^{\theta}$

$\Rightarrow 1-F_X(x) = \left(\frac{x_0}{x}\right)^{\theta}$

$\Rightarrow \ln\left[1-F_X(x)\right] = \theta\ln x_0 - \theta\ln x$

$\Rightarrow \ln(P[X>x]) = (\theta\ln x_0) + (-\theta)\ln x = a + b(\ln x)$.

Here $\ln(P[X>x])$ is a linear function of $\ln x$ with a negative slope.